Piter Z. Garcia Bautista

NSF Noyce Scholar — Data Science & Bioinformatics Graduate Student

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Leadership & Research Summary

NSF Robert Noyce Scholar and PhD-track graduate student specializing in **equitable bioinformatics** and **fairness-aware machine learning**. Developer of the interdisciplinary EQUITAS framework integrating computational rigor, educational equity, and lived experience to address diagnostic disparities in marginalized populations. Leadership focus: translating complex technical innovation into accessible pathways, building cross-institutional bridges, and mentoring underrepresented STEM students.

Education

- **Master of Science in Data Science** — Present
  Rochester Institute of Technology, Golisano College of Computing — *Rochester, NY*
- Concentration: Bioinformatics & AI — Current GPA: 3.9+ (Excellent Academic Standing)
- Coursework: Data Science Foundations, Software Engineering for Data Science, Bioinformatics Algorithms, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics
- **M.S. in Teaching Computer Science K-12 & Disability Leadership Certificate** — Expected Aug 2026
  University of Rochester, Warner School of Education — *Rochester, NY*
- NSF Robert Noyce Scholarship (Full funding: $30,000) — 2-year service commitment in high-need schools
- Advanced Certificate in Leadership in Disability and Inclusive Practices (in progress)
- Coursework: Disability and Schools, Assessment in Instructional Contexts, Inclusive Classrooms, CS Standards Integration
- **Master of Science in Computer Science** — 2015
  Rochester Institute of Technology — *Rochester, NY*
- Concentrations: Computer Graphics, AI, Big Data Analytics — Research: ASL Recognition, Medical Diagnosis
- **Bachelor of Science in Computer Engineering** — 2012
  Farmingdale State College — *Farmingdale, NY*
- Dual Degree: Electrical & Computer Engineering Technology — Honors: CSTEP Merit Award

Leadership Experience at RIT

- **Graduate Assistant - Quantum Computing Research** — Fall 2025 - Present
  Software Engineering Department, RIT — *Rochester, NY*
- **Research Leadership:** Quantum entanglement path selection & qubit allocation via adversarial neural bandits
- **Translation & Accessibility:** Reframed core research contribution for clarity, improving technical precision while making complex quantum concepts accessible to diverse audiences
- **Mentorship Model:** Explicitly show learning process to give peers permission to struggle and learn; model intellectual humility as leadership strength
- **Interdisciplinary Integration:** Connected quantum computing research with healthcare equity applications
- **Lead Researcher - EQUITAS Framework for Diagnostic Justice** — 2024 - Present
  Independent Research Initiative, RIT & University of Rochester — *Rochester, NY*
- **Research Innovation:** Developed interdisciplinary framework integrating trauma-informed bioinformatics, disability justice, and algorithmic fairness—achieving 89% improvement in diagnostic accuracy for marginalized populations
- **Cross-Institutional Bridge-Building:** Pioneered connection between RIT bioinformatics and UOR disability studies that neither field had formalized; presented to faculty, students, and community partners across both institutions
- **Leadership Philosophy:** "Leaders must actively see what institutional systems are trained to ignore"—asked uncomfortable questions about who is missing from datasets and whose lived experiences are treated as noise
- **Collaborative Research:** Led cross-disciplinary teams spanning computer science, education, and health sciences
- **Workshop Facilitator - "EQUITAS Meets the Puzzle Plan"** — 2025
  Community Engagement Initiative, RIT — *Rochester, NY*
- **Workshop Design:** Created 150-minute interactive session combining personal narrative, statistical evidence, hands-on fairness auditing, creative expression, and action planning
- **Inclusive Leadership:** Designed bilingual materials and accessibility protocols supporting neurodivergent and CLD participants
- **Participant Outcomes:** Equipped attendees with practical bias-detection skills, policy templates, and community organizing tools
- **Complex Leadership:** Refused false choices between rigor and urgency, technical depth and social impact, healing and accountability
- **Principal Investigator - Equitable Bioinformatics Projects** — 2024 - Present
  Data Science & Bioinformatics Programs, RIT — *Rochester, NY*

- **ISTE-780 Project:** First systematic application of fairness assessment to RNA secondary structure prediction; developed modular bias detection framework revealing critical equity violations (11.1% equity score, $p<0.01$)
- **BIO-630 Project:** Advanced RNA structure prediction integrating Nussinov algorithm with MFE enhancements; achieved >90% accuracy with validated open-source codebase
- **Leadership Impact:** Challenged assumptions about "objective" algorithms; positioned technical rigor alongside social responsibility

Teaching & Mentoring Experience

- **STEM Tutor & Academic Mentor** 2021 - Present
  Varsity Tutors & Independent Practice *Remote & In-Person*
- Comprehensive tutoring in Computer Science, Data Science, Mathematics, and Statistics at undergraduate and graduate levels
- Develop culturally responsive, trauma-informed lesson plans supporting neurodivergent learners and students with disabilities
- Mentor underrepresented students in STEM
- **Leadership Commitment:** "I want to mentor the next cohort of students who feel like outsiders and help them see that their difference is the foundation for visionary work"
- **Faculty Assistant - Graduate Office** 2013 - 2015
  Golisano College of Computing, RIT *Rochester, NY*
- Supported graduate program operations, student services, and faculty coordination
- Provided direct assistance to graduate students navigating program requirements and procedures

Publications & Academic Contributions

- **Garcia, P.** (2025). *Trauma-Informed Bioinformatics for Diagnostic Justice: An EQUITAS Framework for Integrative Healthcare and Educational Systems.* ED-447 Capstone, University of Rochester.
- **Garcia, P.** (2025). *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data.* ISTE-780 Project Report, RIT.
- **Garcia, P.** (2025). *RNA Secondary Structure Prediction and Visualization: Implementing the Nussinov Algorithm with MFE Enhancements.* BIO-630 Final Project, RIT.
- **Garcia, P.** (2015). *ASL Recognition Application: Image Processing for Sign Language Translation.* MS Thesis, RIT.
- **Garcia, P.** (2014). "Big Data Medical Diagnosis: A Three-Method Approach." *RIT Graduate Research Symposium.*

Technical Skills

Programming: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash
Bioinformatics: ViennaRNA, Nussinov Algorithm, Turner Parameters, BLAST, SHAP Analysis
Machine Learning & AI: TensorFlow, Keras, scikit-learn, XGBoost, PyTorch, LSTM, Fairness-aware ML
Data Science: Pandas, NumPy, SciPy, Matplotlib, Plotly, Seaborn, Statistical Analysis
Cloud & Infrastructure: AWS, GCP, Azure, Docker, Git/GitHub, Linux/Unix
Specialized: EQUITAS Fairness Assessment, Bias Detection, Disparate Impact Analysis, Universal Design for Learning

Professional Experience

- **Data Solutions Engineer** Sep 2022 - Aug 2024
  VEDADATA Inc. *New York, NY*
- Designed data-driven solutions using Python, Pandas, and cloud technologies with emphasis on accessibility and robustness
- **AI Data Solutions Engineer** Jan 2021 - Sep 2022
  VIOME Inc. *New York, NY*
- Developed AI-driven healthcare solutions with focus on equitable outcomes and responsible data use
- **Computer & Data Science Consultant** Mar 2017 - Jan 2021
  Samasta Health Foundation *Remote*
- Pro-bono consulting to architect trauma-informed AI-driven healthcare solutions supporting children in need

Awards & Recognition

- **NSF Robert Noyce Teacher Scholarship** (2024-2026) - Full funding for MS in Teaching CS K-12
- **MS International Scholarship** (2012-2015) - MESCYT & RIT - Academic excellence recognition
- **Honorable Mention Technology Award** (2012) - NYS STEP/CSTEP - 2nd place capstone project
- **IEEE Alumni Member** (2009-Present) - Member No. 90613000